

NOVA Agricultural Disease Dataset EDA Report

Research Verification & Validation Artifact for B.Tech Thesis

1. Executive Summary

This PDF document presents the publication-quality Exploratory Data Analysis (EDA) of the processed dataset used to train the NOVA v1.0 neural classification models. The dataset features **5667 images** across **36 classes**. The class distribution inequality Gini coefficient is calculated at **0.1772**.

2. General Metrics Table

Metric	Value	Notes
Total Images	5667	Balanced classes
Total Classes	36	Capped to 200 samples
Imbalance Ratio	3.51	Max size / Min size
Gini Index	0.1772	0 represents perfect equality
Mean Image Size	180.51 KB	RGB image format
Shannon Diversity	3.5152	Shannon entropy

3. Dataset Split Stratification

The dataset split configuration follows a standard 80/10/10 ratio:

- **Training Split (80%)**: 4088 images
- **Validation Split (10%)**: 1021 images
- **Testing Split (10%)**: 558 images

Zero leakage check: Verified no test image filenames overlap with train/val sets.

4. Publication-Quality Figures Reference

All figures supporting this analysis are compiled in high-resolution 300 DPI under the figures/ directory:

- **Figure 1**: Dataset Pipeline Diagram
- **Figure 2**: Balanced Class Distribution Chart
- **Figure 3**: Resolution and Aspect Ratio Clusters
- **Figure 4**: RGB Average Color Histograms
- **Figure 5**: Augmentation Transformation Pipeline